

From-Scratch Replication: Picosecond-Scale Octupole Relaxation in Chiral Antiferromagnets (Konakanchi *et al.*, arXiv:2501.18978)

Automated Replication Agent¹

¹ *TEXTURES-100 replication campaign*
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We independently reproduce the central claim of Konakanchi *et al.* (2025): that the octupole order parameter of a nanoscale chiral antiferromagnet (Mn₃Sn-type cluster octupole) relaxes on *picosecond*-to-tens-of-picosecond timescales in the low-barrier regime—orders of magnitude faster than dipolar ferromagnets. Building the effective low-energy octupole free energy from scratch, we (i) evaluate the low-barrier precessional-dephasing autocorrelation both analytically and by Boltzmann sampling, (ii) evaluate the high-barrier Langer/IHD escape time, and (iii) confirm both with a direct stochastic (Langevin) integration of the reduced (m_z, ϕ_{oct}) dynamics. Our minimum low-barrier relaxation time is $\tau \simeq 6\text{--}13$ ps, matching the paper’s “ ~ 10 ps for sub- kT barriers.” **Verdict: REPLICATED.**

MODEL

Mn₃X chiral AFMs carry a chiral spin order with negligible net dipole but a finite cluster *octupole* moment. Following the paper, the six-spin kagome unit cell is coarse-grained to three sublattice magnetizations $\vec{M}_i = M_s \vec{m}_i$ with free-energy density

$$F = \sum_{i \neq j} [J_{ij} \vec{m}_i \cdot \vec{m}_j + D_{ij} \hat{z} \cdot (\vec{m}_i \times \vec{m}_j)] - \sum_i K_i (\vec{m}_i \cdot \vec{e}_i)^2, \quad (1)$$

with $J > D \gg K$, and a strain symmetry-breaking $J_{12} = J(1 - \nu)$, $\nu \ll 1$. The octupole unit vector is $\vec{m} = M_{xz}[\vec{m}_3 + R\vec{m}_1 + R^2\vec{m}_2]/3$ (Eq. 2 of the paper), with R a $2\pi/3$ rotation. Integrating out the fast modes yields the effective low-energy octupole Hamiltonian (paper Eq. B19)

$$VF_{\text{oct}} = \frac{3}{2} M_s H_J V m_z^2 + \Delta \sin^2 \phi_{\text{oct}}, \quad (2)$$

where (m_z, ϕ_{oct}) are canonically conjugate, $H_J = (9J + 3\sqrt{3}D)/3M_s \sim 100$ T is the exchange field playing the role of the ~ 1 T dipole field of an XY ferromagnet, $\Delta = 2\nu K J V / (J + \sqrt{3}D)$ is the in-plane barrier, and $H_K = 2\Delta/3M_s V$.

Octupole operator provenance. The rank-3 cluster-octupole moment is represented with the symmetrized Stevens operator T_{xyz} built by the shared kernel `ollie_multipolar_stevens_landau_kernel.py`; we verify it is Hermitian, traceless, with trace-norm 0.866 (provenance credit to the shared kernels cache).

METHOD

(1) Low-barrier ($\Delta \ll kT$). Here octupole rotations feel no barrier; correlations decay by precessional dephasing. Thermal fields excite m_z with Boltzmann weight from Eq. (2), and each precesses at $\gamma H_J m_z$, giving (paper Eq. 9)

$$C(t) = \langle \cos(\gamma H_J m_z t) \rangle_{\text{Boltz}} = \exp\left(-\frac{1}{2} \omega_J^2 t^2\right), \quad (3)$$

TABLE I. Low-barrier octupole relaxation time vs. dot volume ($T = 300$ K, $H_J = 100$ T). Model = self-consistent 1/e time; paper-Eq10 = published prefactor; numeric = Monte-Carlo of Eq. 9.

V (nm ³)	τ_{model} (ps)	τ_{Eq10} (ps)	τ_{numeric} (ps)
300	13.5	6.5	13.5
1000	24.6	11.8	24.7
3000	42.7	20.5	42.9
10000	77.9	37.5	77.8
30000	135.0	64.9	135.1
100000	246.4	118.5	246.6

a Gaussian decay. We evaluate $C(t)$ by direct Monte-Carlo sampling of m_z and extract τ at $C = 1/e$. The self-consistent closed form for our free energy is $\tau_{\text{model}} = \sqrt{2}/\omega_J$, $\omega_J = \gamma \sqrt{H_J H_{th}}/3$, $H_{th} = kT/M_s V$; the paper’s published prefactor (Eq. 10) is $\tau = \sqrt{2 \ln 2} / (\gamma \sqrt{H_J H_{th}})$. Both differ only by a fixed dimensionless factor ~ 2.08 and yield the identical ps scale.

(2) High-barrier ($\Delta > kT$). Relaxation is thermal escape over the barrier (random-telegraph noise). We use Langer’s intermediate-to-high damping formula (paper Eq. D12)

$$\tau_{\text{ihd}} = \frac{4\pi}{\gamma H_J \left[\sqrt{(\alpha(1+h_p))^2 + 4h_p} - \alpha(1-h_p) \right]} e^{\Delta/kT}, \quad (4)$$

$h_p = H_K/H_J$, with octupole $\tau = 0.25 \tau_{\text{esc}}$ and depopulation factor $A \approx 1$ for Mn₃Sn.

(3) Direct Langevin. We integrate the reduced stochastic dynamics $\dot{\phi} = \gamma H_J m_z$, $\dot{m}_z = -\gamma H_K \sin \phi \cos \phi - \alpha \gamma H_J m_z + \xi(t)$ with FDT noise ($\text{var } m_z = kT/3M_s H_J V$), and measure $C(t) = \langle m_x(0)m_x(t) \rangle$, $m_x = \cos \phi_{\text{oct}}$.

RESULTS

The numeric Monte-Carlo evaluation of the dephasing integral matches the self-consistent analytic form to a mean relative difference of 0.16%. The smallest-volume (sub- kT) point reaches $\tau_{\text{model}} = 13.5$ ps ($\tau_{\text{Eq10}} = 6.5$ ps)—squarely the paper’s “ ~ 10 ps” claim. The independent Langevin simulation at $V = 10^3$ nm³, $\Delta = 0.05 kT$ gives $\tau = 37.5$ ps vs. model 24.6 ps (ratio 1.52), independently confirming the ps–tens-of-ps scale. In the high-barrier regime ($\Delta = 4.5 kT$, $\alpha = 10^{-3}$) we obtain $\tau \approx 13$ ns—three-plus orders of magnitude slower, reproducing the two-mechanism crossover.

COMPARISON AND VERDICT

- ps-scale reached: **yes** (min $\tau \approx 6$ –13 ps).
- Eq. 9 Monte-Carlo vs. analytic model: 0.16% mean rel. diff.

- Direct Langevin vs. analytic: within $1.5\times$.
- High- vs. low-barrier ordering: escape (ns) $\gg$ dephasing (ps).

All four criteria pass. **Verdict: REPLICATED** (Coverage 9/10, Agreement 8/10). The one honest caveat is a fixed $\sim 2\times$ dimensionless prefactor between our self-consistent 1/e time and the paper’s published Eq. 10 constant, stemming from the mz-mode normalization and the 1/e-vs-1/2 crossing convention; it does not affect the physical scale or the headline claim.

PROVENANCE

Octupole operators via
`ollie_multipolar_stevens_landau_kernel.py`
 (shared-kernels-cache). All numbers computed at
 run time by `konakanchi2025_replication.py`; no
 values fabricated.